A SUM A DAY

MATHS CLUB

KENYA CERTIFICATE OF BASIC EDUCATION (KCBE) 2026

ESSENTIAL MATHEMATICS

GRADE 10

TERM 2 – 2026

DOSE 1 | ***Time: 30 minutes***

STUDENT DETAILS

Name: ..... **Stream:** ..... **School Code:** .....

SECTION A (10 Marks)

*Answer **all** the questions in this section in the spaces provided.*

1. Classify the following numbers as odd, prime or composite: **(3 marks)**

13, 15, 17, 22, 29

Odd	
Prime	
Composite	

2. A fuel tank whose capacity is 40.5 litres is $\frac{3}{4}$ full. After some time, 19.4 litres were used. How many more litres will be needed to fill the tank? **(3 marks)**

3. Factorise the following quadratic expressions:

(a) $4x^2 - 12x + 9$ **(2 marks)**

(b) $25x^2 + 25x + 4$ **(2 marks)**

SECTION B (10 Marks)

*Answer **all** the questions in this section in the spaces provided.*

4. Apply the laws of indices to work out the following computations:

(a) $\dfrac{243^3 \times 9^2}{729 \div 27}$ **(2 marks)**

(b) $x^7y^8 \div x^5y^3$ **(2 marks)**

5. Two similar triangular tiles have corresponding shortest sides of 6 cm and 15 cm. The other sides of the smaller tile are 8 cm and 11 cm.

(a) Find the linear scale factor of the smaller tile to the larger tile. **(2 marks)**

(b) Find the other two side lengths of the larger tile. **(2 marks)**

(c) Find the ratio of their areas. **(2 marks)**

END OF THE EXAMINATION PAPER

A SUM A DAY

KENYA CERTIFICATE OF BASIC EDUCATION (KCBE) 2026
CORE / ESSENTIAL MATHEMATICS
GRADE 10
TERM 2 – 2026

DOSE 2 | ***Time: 30 minutes***

STUDENT DETAILS

Name: **Stream:** **School Code:**

TOTAL: 15 MARKS

Answer all the questions in the spaces provided.

1. Use logarithm tables to evaluate: **(4 marks)**

$$\sqrt[3]{\frac{0.08472 \times 6.315}{12.48}}$$

2. A delivery van travels 180 km at an average speed of x km/h. If the speed were increased by 15 km/h, the journey would take 1 hour less.

(a) Write an expression for the time taken at:

(i) the original speed; **(1 mark)**

(ii) the increased speed. **(1 mark)**

(b) Form a quadratic equation in x. **(1 mark)**

(c) Solve the equation and determine the original speed of the van. **(2 marks)**

3. A rectangular poultry run has an area of 280 m². Its length is 6 m greater than its width. Form a quadratic equation and hence determine the dimensions of the run. **(3 marks)**

4. Amina is 5 years older than her brother. Three years ago, the product of their ages was 90. Form a quadratic equation and determine their present ages. **(3 marks)**

END OF THE EXAMINATION PAPER

A SUM A DAY

KENYA CERTIFICATE OF BASIC EDUCATION (KCBE) 2026
CORE / ESSENTIAL MATHEMATICS
GRADE 10
TERM 2 – 2026

DOSE 3	*Time: 30 minutes*

STUDENT DETAILS

Name:	Stream:	School Code:

Instructions: Answer all questions and show all working. Use one bold square to represent one unit on the graph paper.

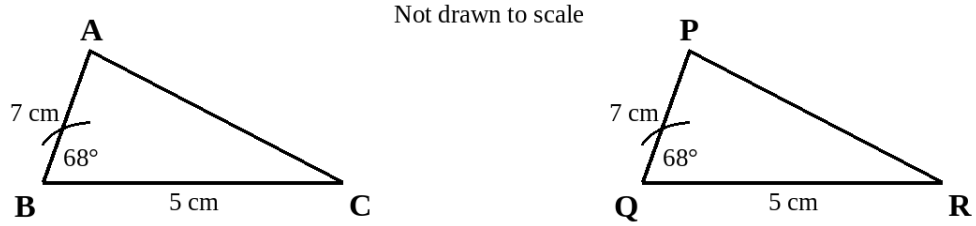
SECTION A: REFLECTION AND CONGRUENCE (10 MARKS)

1. Triangle PQR has vertices P(-4, 1), Q(-1, 3) and R(2, 1). Reflect it in the line y = -x, y= x,x=0 and state the coordinates of P′, Q′ and R′. **(3 marks)**

2. Under a reflection, M(-3, 5) maps onto M′(7, 1). Determine the equation of the mirror line. **(3 marks)**

3. Triangles ABC and PQR satisfy AB = PQ = 7 cm, BC = QR = 5 cm and ∠ABC = ∠PQR = 68°.

Not drawn to scale



(a) State the congruence test that proves triangle ABC is congruent to triangle PQR. Give a reason. **(2 marks)**

(b) If AC = 8.4 cm and ∠ACB = 47°, determine PR and ∠PRQ. **(2 marks)**

SECTION B: REFLECTION ON THE CARTESIAN PLANE (10 MARKS)

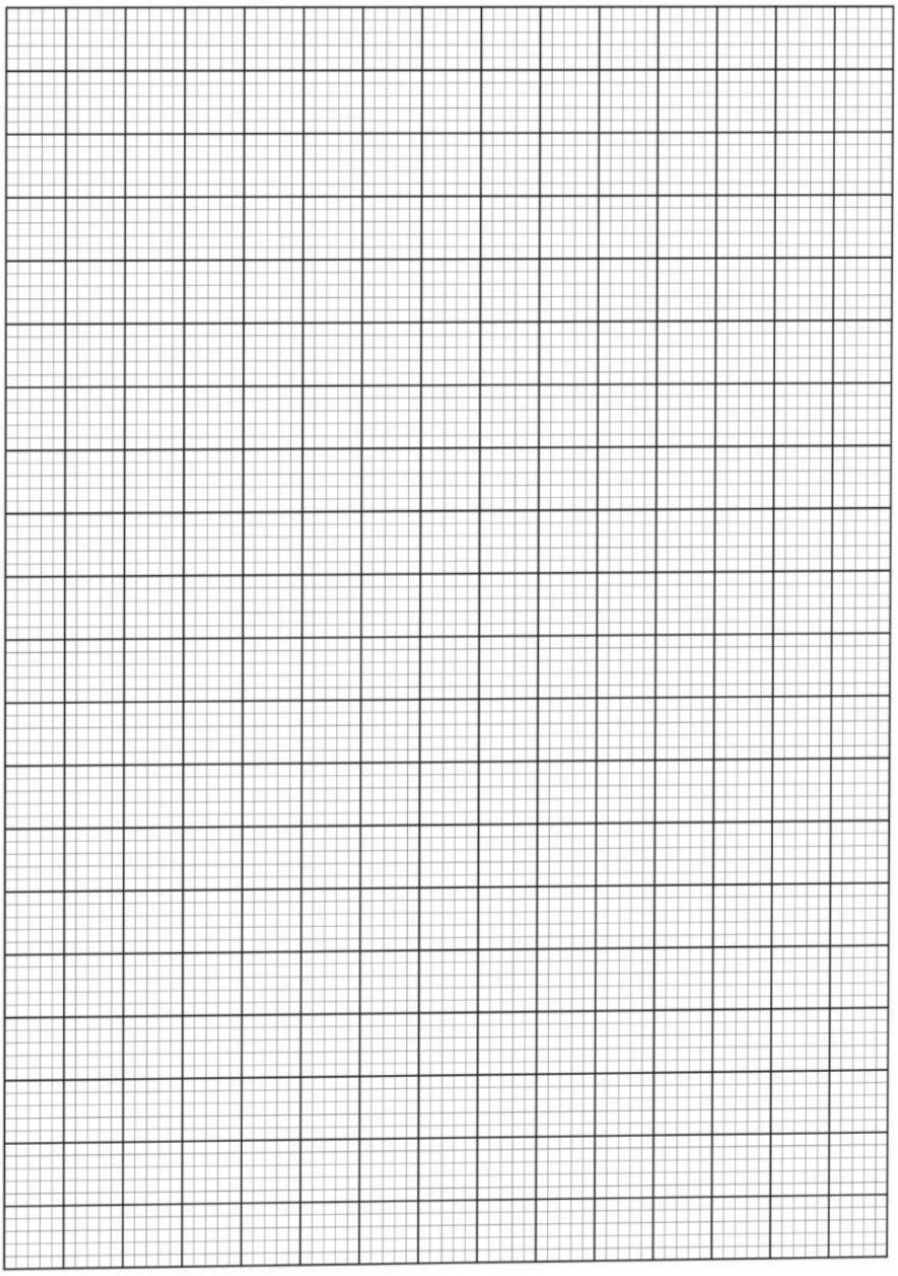
4. (i) The coordinates of triangle ABC are A(2, 1), B(4, 1) and C(3, 3). Triangle ABC is reflected to form triangle A′B′C′, where A′(6, 5), B′(6, 3) and C′(4, 4).

(a) On the graph paper provided, draw suitable Cartesian axes and plot triangle ABC and its image A′B′C′. **(2 marks)**

(b) Describe fully the transformation that maps triangle ABC onto triangle A′B′C′. **(3 marks)**

(c) Reflect triangle A′B′C′ in the line y = x to obtain A″B″C″. On the same grid, draw A″B″C″ and the line y = x. **(3 marks)**

(ii) Point A(2, 4) is reflected onto A′(0, 8). Determine the equation of the mirror line that maps A onto A′. **(2 marks)**



A SUM A DAY

KENYA CERTIFICATE OF BASIC EDUCATION (KCBE) 2028
CORE/ESSENTIAL MATHEMATICS
GRADE 10
AUGUST 2026

DOSE 4 | **Time: 30 minutes**

STUDENT DETAILS

Name: _____ Stream: _____ School Code: _____

SECTION A: INDICES (10 MARKS)

1. Simplify using the laws of indices. **(3 marks)**

(a) Express the following as a single power of 3. **(1 mark)**

$$27^2 \times 9^{-1} \div 3^{-2}$$

(b) Simplify the following and give the answer using positive indices only. **(2 marks)**

$$16a^{-3}b^4 \div (2^{-1}a^2b^{-2})$$

2. A backup server triples stored file-copies after every security cycle. It starts with 2×10^4 file-copies. **(3 marks)**

(a) Write and evaluate the number of file-copies after 5 cycles in standard form. **(2 marks)**

(b) If the server limit is 1×10^7 file-copies, determine the unused capacity. **(1 mark)**

3. Apply fractional indices and exponential equations. **(4 marks)**

(a) Evaluate the following. **(2 marks)**

$$27^{\frac{2}{3}} \times 16^{\frac{-1}{2}}$$

(b) Two population models give equal predictions when the equation below holds. Find x. **(2 marks)**

$$2^{3x+1} = 16^{x-1}$$

SECTION B: LOGARITHMS (10 MARKS)

4. Use the power law of indices. **(3 marks)**

(a) Simplify the expression below in index form. **(1 mark)**

$$(9^2)^3$$

(b) Simplify the expression below. **(1 mark)**

$$3(y^4)^2$$

(c) Simplify the expression below. **(1 mark)**

$$(a^2b^3)^2$$

5. Apply the laws of indices to work out the computation below. Give your answer as a power of 7 and then evaluate it. **(3 marks)**

$$\frac{343 \times 49^2}{7}$$

6. Use indices in the notice-board arrangement. **(4 marks)**

(a) A display board has 2^4 rows with 2^4 cards in each row. Express the total number of cards in index form and then evaluate it. **(2 marks)**

(b) The cards are shared equally among 2^3 learner groups. Find the number of cards each group receives in index form and then evaluate it. **(2 marks)**

END OF THE EXAMINATION PAPER

A SUM A DAY

KENYA CERTIFICATE OF BASIC EDUCATION (KCBE) 2026
CORE / ESSENTIAL MATHEMATICS
GRADE 10
TERM 2 – 2026

DOSE 5 | ***Time: 30 minutes***

STUDENT DETAILS

Name: _____ **Stream:** _____ **School Code:** _____

Instructions: Answer all questions and show all working.

SECTION A: SIMILARITY AND ENLARGEMENT (10 MARKS)

1. A rectangular school garden is represented on a plan by a rectangle 7.5 cm long and 4.5 cm wide. The actual garden is 30 m long.

(a) Determine the linear scale factor of the plan to the actual garden, in the form 1 : n. **(2 marks)**

(b) Calculate the actual width of the garden in metres. **(1 mark)**

2. Two similar triangular solar-panel frames have corresponding base lengths of 1.2 m and 1.8 m. The area of the smaller frame is 0.96 m². Calculate the area of the larger frame. **(3 marks)**

3. Two geometrically similar rainwater tanks have capacities of 864 litres and 2 916 litres. The smaller tank is 80 cm high.

(a) Express the volume scale factor, smaller tank : larger tank, in its simplest form. **(1 mark)**

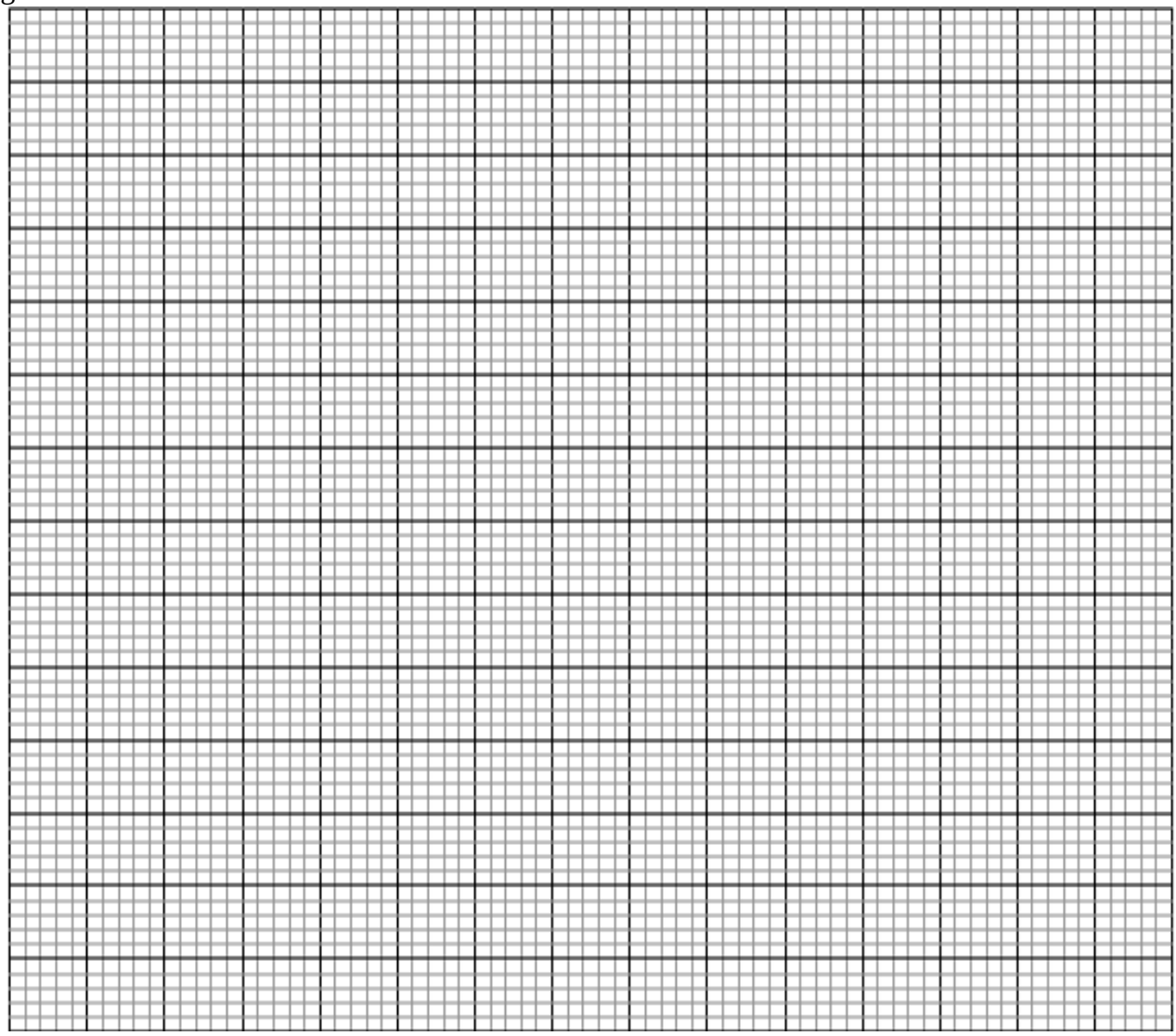
(b) Determine the corresponding linear scale factor, smaller tank : larger tank. Show your reasoning. **(2 marks)**

(c) Hence calculate the height of the larger tank. **(1 mark)**

SECTION A TOTAL: 10 MARKS

SECTION B: ENLARGEMENT AND REFLECTION (10 MARKS)

4. (i) On a graph, plot the points A(1, 1), B(2, 2), C(5, 1) and D(4, 0). Join the points with straight lines. **(2 marks)**



(ii) On the same axes, locate and plot A′, B′, C′ and D′, the images of A, B, C and D under an enlargement with centre at the origin and scale factor –2. **(2 marks)**

(iii) Draw quadrilateral A′B′C′D′ and state the coordinates of its vertices. **(2 marks)**

(iv) Locate and plot A″, B″, C″ and D″, the images of A, B, C and D under reflection in the line y = –x. Hence draw quadrilateral A″B″C″D″. **(3 marks)**

(v) State the coordinates of A″, B″, C″ and D″. **(1 mark)**

TOTAL: 20 MARKS

A SUM A DAY

KENYA CERTIFICATE OF BASIC EDUCATION (KCBE) 2028
CORE / ESSENTIAL MATHEMATICS
GRADE 10
AUGUST 2026

DOSE 6 | ***Time: 30 minutes***

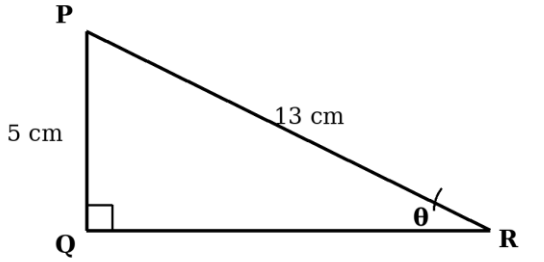
STUDENT DETAILS

Name: _____ **Stream:** _____ **School Code:** _____

Instructions: Answer all questions and show all working. Use a calculator where necessary.

SECTION A: TRIGONOMETRIC RATIOS AND EQUATIONS (10 MARKS)

1. The diagram shows right-angled triangle PQR, where PQ = 5 cm, PR = 13 cm and ∠PQR = 90°.



Not drawn to scale

(a) Calculate the length QR. **(1 mark)**

(b) Hence determine tan θ. **(1 mark)**

(c) Calculate θ, correct to the nearest degree. **(1 mark)**

2. Solve for x if cos (2x + 10°) = sin (3x – 5°), given that both angles are acute. **(3 marks)**

3. A 6.5 m support cable is attached to a vertical communication mast and anchored 2.5 m from the base of the mast.

(a) Calculate the height above the ground at which the cable is attached to the mast. **(2 marks)**

(b) Determine the angle that the cable makes with the horizontal ground, correct to the nearest degree. **(2 marks)**

SECTION A TOTAL: 10 MARKS

SECTION B: APPLICATION OF TRIGONOMETRY (10 MARKS)

4. Amina, whose eye level is 1.6 m above the ground, stands 14 m from the base of a vertical flagpole. The angle of elevation of the top of the flagpole from her eye level is 48°. Calculate the height of the flagpole, correct to 1 decimal place. **(3 marks)**

5. From the top of a 24 m high building, the angle of depression of a car on level ground is 38°. Calculate the horizontal distance of the car from the base of the building, correct to 1 decimal place. **(3 marks)**

6. An observer at the top of a 60 m high lighthouse sees two boats on the same side of the lighthouse and in the same straight line. The angles of depression of the nearer and farther boats are 52° and 35°, respectively. Calculate the distance between the two boats, correct to 1 decimal place. **(4 marks)**

TOTAL: 20 MARKS

A SUM A DAY

KENYA CERTIFICATE OF BASIC EDUCATION (KCBE) 2028

CORE / ESSENTIAL MATHEMATICS

GRADE 10

AUGUST 2026

DOSE 7 | ***Time: 30 minutes***

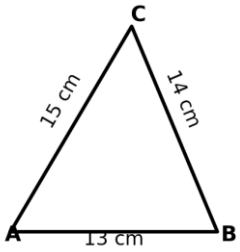
STUDENT DETAILS

Name: _____ **Stream:** _____ **School Code:** _____

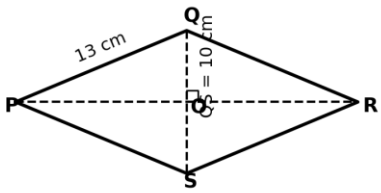
Instructions: Answer all questions and show all working. Use a calculator where necessary.

SECTION A: AREA OF POLYGONS (10 MARKS)

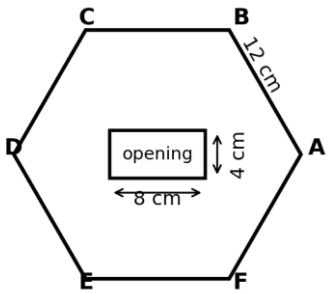
1. The triangular metal plate ABC has sides AB = 13 cm, BC = 14 cm and CA = 15 cm, as shown. Using Determine the area of the plate. **(3 marks)**



2. In rhombus PQRS, PQ = 13 cm and diagonal QS = 10 cm. The diagonals intersect at O. Without measuring the diagram, calculate the area of the rhombus. **(3 marks)**



3. A regular hexagonal metal sheet ABCDEF has side length 12 cm. A rectangular opening measuring 8 cm by 4 cm is cut from its centre, as shown. Calculate the area of metal remaining, correct to 1 decimal place. **(4 marks)**



SECTION A TOTAL: 10 MARKS

SECTION B: APPLICATION OF AREA OF POLYGONS (10 MARKS)

4. A triangular flower bed has two adjacent sides of 18 m and 24 m with an included angle of 52°. Turf is supplied in rolls that each cover 6 m². Determine the minimum number of rolls required to cover the flower bed completely. **(3 marks)**

5. A rhombus-shaped courtyard has diagonals of 18 m and 24 m. Square paving blocks cover 0.45 m² each. The contractor allows an additional 5% of the exact number of blocks for breakages. Determine the total number of blocks that should be ordered. **(3 marks)**

6. A regular hexagonal playground has side length 10 m. A smaller regular hexagonal sandpit of side length 4 m is constructed at its centre. The remaining area is grassed at KSh 650 per square metre. Calculate the total cost of grassing, to the nearest shilling. **(4 marks)**

TOTAL: 20 MARKS

End of the examination paper

A SUM A DAY

KENYA CERTIFICATE OF BASIC EDUCATION (KCBE) 2028

CORE / ESSENTIAL MATHEMATICS

GRADE 10

TERM 2 - 2026

DOSE 7B | ***Time: 30 minutes***

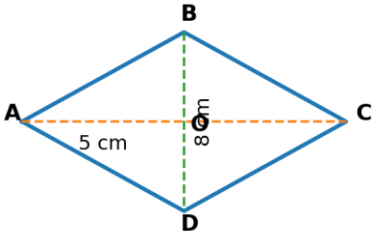
STUDENT DETAILS

Name: _____ **Stream:** _____ **School Code:** _____

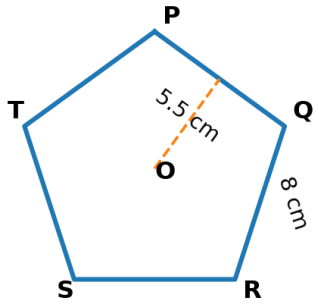
Instructions: Answer all questions and show all working. Use a calculator where necessary.

SECTION A: AREA OF POLYGONS (10 MARKS)

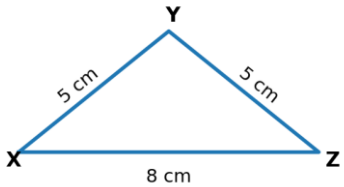
1. In rhombus ABCD, diagonals AC and BD intersect at O. AO = 5 cm and BD = 8 cm. Calculate the area of the rhombus. **(3 marks)**



2. A regular pentagon PQRST has side length 8 cm. The perpendicular distance from its centre O to each side is 5.5 cm. Calculate its area by dividing it into five congruent triangles. **(3 marks)**



3. A triangular card XYZ has side lengths XY = 5 cm, YZ = 5 cm and XZ = 8 cm. Use Heron's formula to calculate its area. **(4 marks)**



SECTION A TOTAL: 10 MARKS

SECTION B: SIMPLE APPLICATIONS OF AREA OF POLYGONS (10 MARKS)

4. A rhombus-shaped window decoration has diagonals 12 cm and 8 cm. Calculate its area. One sheet of coloured paper covers 24 cm². Determine the number of sheets required. **(3 marks)**

5. A regular pentagonal flower bed has side length 6 m and a perpendicular distance of 4 m from its centre to each side. Calculate its area. One bag of manure covers 12 m². Determine the minimum number of bags required. **(3 marks)**

6. A triangular school flag has sides 5 m, 5 m and 6 m. Use Heron's formula to calculate its area. The material costs KSh 200 per square metre. Calculate the total cost of the material. **(4 marks)**

TOTAL: 20 MARKS

End of the examination paper

A SUM A DAY

KENYA CERTIFICATE OF BASIC EDUCATION (KCBE) 2028

CORE / ESSENTIAL MATHEMATICS

GRADE 10

TERM 2 - 2026

DOSE 9 | ***Time: 30 minutes***

STUDENT DETAILS

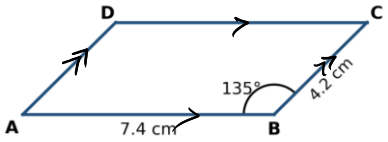
Name: _____ **Stream:** _____ **School Code:** _____

Instructions: Answer all questions and show all working. Use a calculator where necessary.

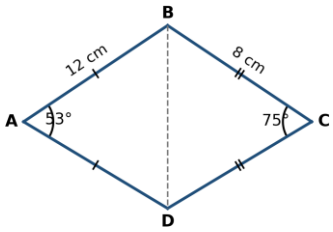
SECTION A: AREA OF POLYGONS (10 MARKS)

1. Find the area of the following figures.

(a) **(2 marks)**

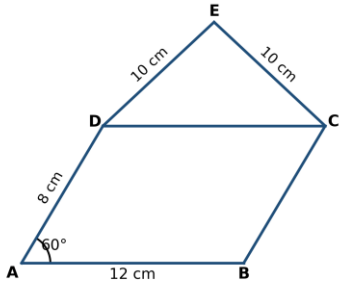


(b) **(2 marks)**



2. A triangular flower bed has a perimeter of 42 m. Two of its sides measure 13 m and 14 m. Determine the area of the flower bed. **(3 marks)**

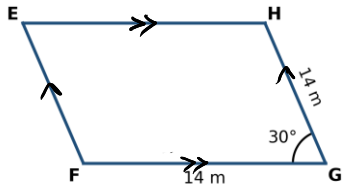
3. The composite metal badge shown below is made by joining an isosceles triangle DCE to parallelogram ABCD along side DC. AB = 12 cm, AD = 8 cm, ∠DAB = 60°, and DE = CE = 10 cm. Calculate the total area of the badge. **(3 marks)**



SECTION A TOTAL: 10 MARKS

SECTION B: APPLICATIONS OF AREA OF POLYGONS (10 MARKS)

4. Mwalimu Daniel owns a farm and intends to make a chicken enclosure as shown below. Each chicken needs 2 m² of space to move around freely. Determine the maximum number of chickens that can fit in the enclosure. **(3 marks)**



5. Kalikwa is making a 12 cm by 10 cm parallelogram-shaped metal plate with an area of 36 cm² for a decorative wall piece. Determine the sizes of the angles of the parallelogram. **(4 marks)**

6. A trampoline cover is designed in the shape of a regular hexagon with each side measuring 1.5 m. The material used for the cover costs KSh 250 per square metre.

(a) Calculate the area of the material used to make the cover. **(2 marks)**

(b) Determine the total cost of the trampoline cover. **(1 marks)**

TOTAL: 20 MARKS

End of the examination paper